

Yakine Audio Learner

Detailed Report Structure (Combined Sprints Version)

End of Studies Project (PFE Report)

Front Matter (Approx. 8-10 pages)

Title Page / Cover Page

Dedication & Acknowledgements

Abstract

Table of Contents, List of Figures, List of Tables, List of Acronyms

General Introduction (Approx. 3-4 pages)

Context and Problematic

Proposed Solution

Objectives and Contributions

Report Organization

1 | Chapter 1: Project Context and State of the Art (Approx. 10 pages)

1.1 Introduction

1.2 Project Context (The Problematic and Target Audience)

1.3 State of the Art (Existing Solutions and Comparative Analysis Table)

1.4 Technological Choices (React Native, Express, PostgreSQL)

1.5 Conclusion

2 | Chapter 2: System Requirements & Global Architecture (Sprint 0) (Approx. 15 pages)

2.1 Introduction

2.2 Methodology (Agile Scrum)

2.3 Requirement Analysis

2.3.1 Identification of Actors (Guest, User, Teacher, Admin)

2.3.2 Functional Requirements (Includes Global Use Case Diagram)

2.3.3 Non-Functional Requirements

2.4 Global Architecture

2.4.1 3-Tier Architecture (Includes Global Architecture Diagram)

2.4.2 Domain Model (Includes ERD / schema.prisma representation)

2.5 Global Product Goals

2.6 Conclusion

3 | Chapter 3: Sprint 1 - Authentication & Security (Approx. 12 pages)

- 3.1 Introduction
- 3.2 Sprint 1 Backlog (User Stories 1.1 to 1.12)
- 3.3 Software Design
 - 3.3.1 Use Case Diagram: Registration & Authentication
 - 3.3.2 Sequence Diagram: JWT Authentication & Session Invalidation
- 3.4 Implementation (Login, Registration, Profile Management)
- 3.5 Conclusion & Retrospective

4 | Chapter 4: Sprints 2 & 3 - Learning Experience & Access Governance (Approx. 25-28 pages)

- 4.1 Introduction
- 4.2 Sprint 2 Backlog (Catalog, Content Management, Access Governance)
- 4.3 Sprint 3 Backlog (Audio Playback, Multilingual Learning, Progress, Offline Mode)
- 4.4 Software Design
 - 4.4.1 Use Case Diagram: Catalog Browsing & Content Management
 - 4.4.2 Access Control Logic: Premium locks and teacher scopes
 - 4.4.3 Sequence Diagram: Audio Playback & Progress Tracking
 - 4.4.4 State Machine Diagram: Download Management
- 4.5 Implementation (Subjects, Chapters, Lessons, Audio Player, Progress Tracking, Offline Features)
- 4.6 Conclusion & Retrospective

5 | Chapter 5: Sprints 4 & 5 - Governance and Content Generation (Approx. 18-20 pages)

- 5.1 Introduction
- 5.2 Sprints Backlog (Admin operations, Feature Flags, Teacher Composer, AI TTS tasks, Bulk Data)
- 5.3 Software Design
 - 5.3.1 Admin & Moderation Use Case Diagram
 - 5.3.2 AI Audio Generation Pipeline Sequence Diagram
- 5.4 Implementation (Admin Dashboard, Moderation, Teacher Composer, AI Tasks)

5.5 Conclusion & Retrospective

General Conclusion (Approx. 2-3 pages)

Summary of Achievements

Limitations

Future Perspectives

Bibliography & Appendices (Approx. 5-8 pages)

Bibliography

Appendix A: Complete Detailed Product Backlog (Sprints 1 through 5)

Appendix B: API Endpoints or Database Schema